SOVEREIGN Game

SOVEREIGN Game Rulebook: A Strategic Simulation Environment for Deep Reinforcement Learning Research

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Note: This is a pilot project intended to promote unusual but meaningful interdisciplinary collaboration between computer science and peace studies. The following is an interesting research by Meta. This rulebook was inspired by the Diplomacy board game. I drafted this rulebook with a help of an AI agent. There could be some ambiguity in the definition (mainly because the state and action of the original Diplomacy game was too huge to simply..., but I tried my best.). If you find any issue or unclear definitions, please report it to me.

- Meta Fundamental AI Research Diplomacy Team (FAIR) et al., Human-level play in the game of Diplomacy by combining language models with strategic reasoning. Science, 378, 1067-1074 (2022). DOI:10.1126/science.ade9097 (PDF (https://noambrown.github.io/papers/22-Science-Diplomacy-TR.pdf))
- Diplomacy package (https://diplomacy.readthedocs.io/en/stable/) : Diplomacy is a strategic board game when you play a country (power) on a map with the goal to conquer at least half to all the supply centers present on the map.

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1. Overview

SOVEREIGN is a deterministic-core, stochastic-politics strategic simulation designed as a Gymnasium-compatible environment for deep reinforcement learning research. It models a three-nation geopolitical conflict in which one nation (the **Invader**) holds a clear military advantage and must decide how to pursue its strategic objectives through a joint military–political action space.

The central research question the environment is designed to probe: *Can a militarily superior agent learn, through experience alone, that invasion is a strategically dominated strategy?*

The environment does not punish invasion by fiat. The cost of aggression emerges organically through the interaction of political legitimacy, economic attrition, occupation drag, and shifting international coalitions.

2. Nations

The game is played with exactly **three nations**. Their roles are fixed at initialization.

Nation	Role	Starting Condition
Invader (I)	The learning agent. Controls the DRL policy.	Hard-power advantage: 2× military units, higher strike capacity
Defender (D)	Rule-based opponent. Responds to Invader actions.	Home-turf advantage: +20% unit effectiveness on own territory
Neutral (N)	Stochastic observer. Posture shifts based on Invader behavior.	Begins at `θ = 0` (true neutral)

2.1 Hard-Power Asymmetry

The Invader begins the game with a deliberate and significant military advantage. This is a **design requirement**, not a coincidence. The thesis of the environment is that a *superior* agent learns that invasion is irrational. If the Invader were weak, the result would be trivially explained by military inferiority.

Default initialization:

```
Invader:   12 ground units,  3 strike units,  supply index = 1.0
Defender:   6 ground units,  1 strike unit,   supply index = 1.0
Neutral:    4 ground units,  0 strike units   (non-combatant at θ = 0)
```

3. The Game Board

The map is a **graph** `G = (V, E)` where nodes are territories and edges are adjacency (movement is permitted only along edges).

3.1 Territories

Each territory `v ∈ V` has three attributes:

Attribute	Type	Description
`controller`	{I, D, N, Contested}	Which nation controls the territory
`resource_value`	Float [0, 1]	Economic output contributed per step if controlled
`strategic_value`	Float [0, 1]	Military significance (affects supply lines)

3.2 Supply Lines

A nation's **supply index** `E ∈ [0, 1]` is computed each step as a function of territory connectivity. Occupied territories that are not contiguous with the Invader's home territory generate *no* resource value and increase occupation cost.

3.3 Default Map

The default map has **9 territories**: 3 home territories (one per nation) and 6 contested territories. You can create one sample topology. Also, you can consider testing the performance in other topologies. (Show at least one map.)

4. State Space

At each timestep `t`, the environment emits a state vector `s_t` comprising the following components.

4.1 State Variables

Variable	Symbol	Type	Description
Territory control map	`M`	Binary matrix |V| x 3	One-hot encoding of controller per territory
Military strength	`U_I`, `U_D`	Integer vector	Unit counts per territory for Invader and Defender
International legitimacy	`L`	**Float [0, 1]**	**Invader's standing in the international community**
Economic supply index	`E`	**Float [0, 1]**	**Invader's economic and logistical health**
Neutral posture	`θ`	**Float [-1, +1]**	**Neutral nation's alignment**
Occupation duration	`t_occ`	Integer ≥ 0	Consecutive steps the Invader has held non-home territory

Bold variables are the soul of the design. They are the mechanisms through which the long-run cost of aggression manifests. Ablating any of them materially changes the agent's optimal policy.

4.2 Legitimacy `L`

`L` represents the Invader's international standing. It begins at `L = 1.0` and decays under aggressive action. It recovers slowly under diplomatic action.

- Governs how the Neutral nation responds to the Invader
- Determines whether sanctions are imposed (threshold event at `θ > 0.6`)
- Terminal collapse condition: `L = 0`

4.3 Neutral Posture `θ`

`θ ∈ [-1, +1]` encodes the Neutral nation's political alignment:

```
θ = -1.0   →  fully allied with Invader (supply routes open, no sanctions)
θ =  0.0   →  true neutral
θ = +1.0   →  fully allied with Defender (sanctions active, military support)
```

See **Section 7** for the full posture shift model.

4.4 Occupation Duration `t_occ`

`t_occ` is a monotonically increasing counter that resets only if the Invader withdraws all units from non-home territory. It drives:

- Per-step occupation cost (linear in `t_occ`)
- Insurgency probability `p(insurgency | t_occ)` — a stochastic event that destroys one Invader unit per occurrence

5. Turn Structure

Each game step proceeds in the following fixed sequence:

```
┌──────────────────────────────────────────────────────────┐
|  STEP t                                                  |
|                                                          |
|  1.  Observe state s_t                                   |
|  2.  Invader selects political action  a_pol             |
|  3.  Invader selects military action   a_mil             |
|  4.  Defender responds (rule-based policy)               |
|  5.  Resolve military outcomes (deterministic)           |
|  6.  Update territory control map M                      |
|  7.  Update L, E, t_occ                                  |
|  8.  Sample Neutral posture shift  Δθ                    |
|  9.  Check threshold events (sanctions, coalition)       |
|  10. Check terminal conditions                           |
|  11. Compute reward  r_t                                 |
|  12. Emit  (s_{t+1}, r_t, done, info)                    |
└──────────────────────────────────────────────────────────┘
```

The political action is committed before the military action. This is a deliberate design constraint. The agent cannot use military force as a substitute for political reasoning within a single step.

6. Action Space

The Invader's action at each step is a **joint action** `a = (a_pol, a_mil)`.

6.1 Political Actions `A_pol`

Action	Effect on `L`	Effect on `θ`	Effect on `E`
`SEEK_ALLIANCE`	+0.01	−0.05 (toward Invader)	—
`IMPOSE_SANCTION`	−0.02	+0.04 (toward Defender)	−0.03 (target)
`ISSUE_THREAT`	−0.03	+0.03	—
`NEGOTIATE`	+0.03	−0.04	—
`DO_NOTHING`	Slow decay if `L < 0.5`	Slow drift if `t_occ > 0`	—

6.2 Military Actions `A_mil`

Action	Territory effect	Effect on `L`	Effect on `t_occ`
`ADVANCE`	Claim adjacent territory	−0.05	+1 per step in non-home
`HOLD`	Maintain current positions	—	+1 per step in non-home
`WITHDRAW`	Cede one contested territory	+0.02	Resets if full withdrawal
`STRIKE`	Destroy one Defender unit	−0.08	+1

7. Neutral Posture System

This section describes the most nuanced mechanism in the environment.

7.1 Drift-Diffusion Model

Neutral posture evolves according to:

```
θ_{t+1} = clip( θ_t  +  μ(s_t, a_t)  +  ε,  -1, +1 )

where  ε ~ N(0, σ²),   σ = 0.02  (default)
```

The **deterministic drift** `μ` encodes the rational response of the international community to observable Invader behavior. The **stochastic noise** `ε` models the irreducible uncertainty of international politics.

7.2 Drift Function

```
μ(s_t, a_t) = + α · (1 − L)          # low legitimacy alienates neutral
            + β · [a_mil = ADVANCE]  # territorial advance shocks posture
            + γ · [a_mil = STRIKE]   # strikes carry heavier cost
```

```
            − δ · [a_pol = NEGOTIATE]# diplomacy pulls posture toward center
            − ε · [a_pol = SEEK_ALLIANCE]  # alliance-seeking reduces drift
            + ζ · (t_occ / T_max)   # prolonged occupation steadily alienates
```

Default coefficients:

Coefficient	Symbol	Default value
Legitimacy coupling	`α`	0.04
Advance shock	`β`	0.05
Strike shock	`γ`	0.10
Negotiate pull	`δ`	0.04
Alliance pull	`ε`	0.03
Occupation drift	`ζ`	0.03

7.3 Threshold Events

When `θ` crosses discrete thresholds, **irreversible** geopolitical events are triggered:

Threshold	Event	Mechanical Effect
`θ > 0.60`	Neutral imposes sanctions	Invader `E` reduced by 0.01/step
`θ > 0.85`	Neutral joins Defender	Defender receives +2 units; Invader `L` −0.10
`θ < −0.60`	Neutral opens supply routes	Invader occupation cost reduced 30%
`θ < −0.85`	Neutral formally allies Invader	Defender `E` reduced; `L` −0.05

> **Design note:** Thresholds are one-directional by default. Once sanctions are imposed, they are not lifted unless `θ` falls back below 0.50 and stays there for 5 consecutive steps. This hysteresis models the political difficulty of reversing international coalitions.

8. Reward Function

The reward at each step is:

```
r_t = r_pos(s_t, a_t)  −  r_neg(s_t, a_t)
```

8.1 Positive Terms

Component	Formula	Description
Territory control	`w_T · Σ resource_value(v) for v controlled by I`	Economic yield of held territory
Resource capture	`w_R · Δ(controlled resources)`	Bonus for newly captured territory

8.2 Negative Terms (per step)

Component	Formula	Description
Occupation cost	`w_O · t_occ / T_max`	Linear cost of sustained occupation
Legitimacy loss	`w_L · (1 − L)`	Penalty proportional to legitimacy deficit
Sanction penalty	`w_S · [θ > 0.6] · (1 − E)`	Active when sanctions threshold crossed
Insurgency event	`w_I · Bernoulli(p(insurgency \ t_occ))`	Stochastic unit destruction

Default weights:

Weight	Default
`w_T`	0.30
`w_R`	0.20
`w_O`	0.25
`w_L`	0.15
`w_S`	0.20
`w_I`	0.10

8.3 Insurgency Probability

```
p(insurgency | t_occ) = 1 − exp(−λ · t_occ)

Default:  λ = 0.05
```

This is a standard reliability-theory hazard function. At `t_occ = 10`, insurgency probability per step is approximately 40%. At `t_occ = 20`, it exceeds 63%.

9. Terminal Conditions

An episode ends when any of the following conditions is met:

Condition	Type	Terminal Reward
`L ≤ 0`	Political collapse	**−50** (large penalty)
All Invader units destroyed	Military defeat	**−30**
Negotiated settlement reached	Diplomatic resolution	**+40** (large bonus)
`t ≥ T_max`	Time limit	`0` (no bonus)
Invader controls all territories	Total conquest	**+10** (intentionally modest)

10. The Core Experimental Protocol

Train the agent under varying parameter regimes and measure whether the peace-dominant policy emerges:

Experiment	`L` active	`t_occ` active	`θ` active	Expected optimal policy
Full model	✓	✓	✓	Negotiate or deter
No legitimacy	✗	✓	✓	Slower invasion
No occupation cost	✓	✗	✓	Partial invasion
No neutral posture	✓	✓	✗	Invasion
Baseline (all off)	✗	✗	✗	Always invade

Also, analyze the impact of different parameter values. (e.g., if the sanction starts earlier, does it change the behavior of the invader?)